

Practicing Data Wrangling with Tree Equity Data

in-class worksheet

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Instructions

In this worksheet, you'll practice writing **tidyverse-style R code** for data wrangling tasks. Each question guides you through filtering, selecting, mutating, and summarizing the `ca_tes` dataset.

Don't stress yourself out over perfect syntax – the goal is to practice what you've learned about data wrangling so far.

Practice Questions

1. Filter for neighborhoods with low tree equity scores

Show only neighborhoods where the `tes` (Tree Equity Score) is **less than 50**.

Hint: Use `filter()`

2. Focus on specific columns of interest

Keep only the following columns: `treecanopy`, `tes`, and `unemplnorm`.

Hint: Use `select()`

3. Calculate percent of tree coverage goal met

Compute a new column that represents **the percent of the tree coverage goal that has been met**.

Hint: Use `mutate()`

- You'll need to figure out how to calculate the value in question — discuss with a partner if unsure!
- Remember that `treecanopy` is the actual tree coverage in an area. `tc_goal` is the ideal tree coverage based on needs.

4. Compute average tree equity score by place

Group the data by `place` (e.g., city or neighborhood), and compute the **average tes score** in each place.

Hint: Use `group_by()` and `summarise()`

5. Which neighborhoods have the greatest tree equity need?

Use only the variables `GEOID`, `place`, and `tes`.

Hint: Use `arrange()`

6. How does tree equity relate to both poverty and health disparity?

In this question, you'll explore the relationship between Tree Equity Score (`tes`), poverty, and health disparity.

- create variable `combined_index` that connects `pctpovnorm` and `health_nor` as a product
- cut data set down to desired variables (`place`, `combined_index`, and `tes`)
- organize data to see which places have highest combined index

Hint: to sort *descending*, you'll need to use `desc()` *within* the `arrange()` function.

